Online Appendix to "Equilibrium Transition from Loss-Leader Competition: How Advertising Restrictions Facilitate Price Coordination in Chilean Pharmaceutical Retail"

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This Online Appendix is for online publication only and accompanies the main text. Cross-references of the form § n, Table n, and equation (n) refer to the main text; appendix material is numbered A–I below.

A Demand-fit decomposition

Goodness-of-fit formulas. The within-nest share prediction used for the diagnostics of §4.5 is

$$\hat{s}_{i|jt} = \frac{\exp\big(\hat{u}_{ij(t)}/(1-\hat{\sigma})\big)}{\sum_{i'\in\mathcal{I}}\exp\big(\hat{u}_{i'j(t)}/(1-\hat{\sigma})\big)}, \qquad \hat{u}_{ij(t)} = \hat{\phi}_{ij} - \hat{\alpha}_{\text{eff}}(t)\, p_{ijt}, \tag{1}$$

with $\hat{\phi}_{ij}$ the within-(chain, drug) fixed effect and $\hat{\alpha}_{\text{eff}}(t) = \hat{\alpha}_0 + \hat{\alpha}_1 D_t$. Only the structural parameters and prices enter the right-hand side, so (1) is a non-circular prediction, not the mechanical contraction-mapping R^2 of Conlon and Gortmaker [2020, p. 32]. Own- and cross-price elasticities use the standard nested-logit formulas

$$\varepsilon^{\text{own}}_{ijj,t} = -\alpha\, p_{ijt}\left[\tfrac{1}{1-\sigma} - \tfrac{\sigma}{1-\sigma}\, s_{i|jt} - s_{ijt}\right], \tag{2}$$

$$\varepsilon^{\text{cross}}_{ijk,t} = \alpha\, p_{kjt}\, s_{kjt}\left[1 + \tfrac{\sigma}{(1-\sigma)\, s_{jt}}\right], \quad i \neq k. \tag{3}$$

The pre-ban share-fit correlation 0.73 reported in Table 3 is below the post-ban value of 0.86. This appendix decomposes the gap. The three relevant sub-regimes are: pre-ban competitive weeks (panel-wide below-cost prevalence ≤ 25%), pre-ban war weeks (> 25% below-cost prevalence), and post-ban weeks. At the headline parameters $(\hat{\alpha}_0, \hat{\alpha}_1, \hat{\sigma}) = (0.103, -0.074, 0.393)$:

Table 1: Below-cost share and implied elasticity

ATC1 class	N drugs	Below-cost share	$\lvert\varepsilon^{\text{own}}\rvert$
H (systemic hormones)	6	0.45	1.16
L (antineoplastics)	2	0.40	0.31
S (sensory organs)	3	0.30	0.19
C (cardiovascular)	38	0.26	1.35
G (genito-urinary)	42	0.23	1.28
A (alimentary tract / metab.)	20	0.13	0.78
D (dermatologicals)	1	0.10	3.47
B (blood / blood-forming)	10	0.10	1.24
N (nervous system)	59	0.07	1.08
R (respiratory)	22	0.06	2.15
M (musculo-skeletal)	14	0.06	1.65
J (anti-infectives, systemic)	5	0.01	3.26

Note: Below-cost share is the fraction of chain-drug-week observations in the pre-ban window with $p_{ijt} < c_{jt}$; $\lvert\varepsilon^{\text{own}}\rvert$ is the class-mean of $\lvert\varepsilon^{\text{own}}_{ijj,t}\rvert$ from (2) at median pre-ban prices. Spearman rank correlation across the 12 classes: $\rho = -0.62$.

Table 2: Share-fit decomposition by regime

Regime	N	Corr$(\hat{s}_{i\mid jt},\, s_{i\mid jt})$
pre-ban competitive	44,421	0.67
pre-ban war	18,564	**0.88**
pre-ban (all)	62,985	0.73
post-ban	41,769	0.86

Note: Pre-ban "war weeks" are pre-ban weeks with panel-wide below-cost prevalence above 25% (28 weeks). Pre-ban "competitive" are the remaining pre-ban weeks.

The pre-ban war weeks fit better than the post-ban regime ($\rho = 0.88$ vs. 0.86). The fit gap is therefore concentrated in the pre-ban competitive weeks, where the static nested-logit struggles to match the strategic dimension of loss-leader behaviour: in those weeks each chain was independently selecting which drugs to push below cost and by how much, drug-by-drug and week-by-week. These are coordinated intertemporal pricing decisions that do not enter the right-hand side of (2); the nested-logit sees them as residuals ξ_{ijt}. Post-ban, the three chains converge to a single coordinated tier and the strategic dimension flattens.

I tested allowing the price coefficient to differ in war weeks, $\alpha_t = \alpha_0 + \alpha_1 D_t + \alpha_2 W_t$ where W_t is the war-week indicator, by a grid search that maximises pre-ban share correlation.

The optimum is $\hat{\alpha}_2 = +0.05$ but the gain in pre-ban ρ is negligible. Removing the regime split entirely and fitting a whole-period $(\hat{\alpha}, \hat{\sigma})$ by share-fit grid search yields $(\hat{\alpha}, \hat{\sigma}) = (0.05, 0.15)$ with whole-period $\rho = 0.80$, marginally above the pooled spec's $\rho = 0.78$. The modest gain confirms that the binding constraint is the within-competitive-week strategic dimension, not the between-regime price coefficient. A fuller resolution would require a dynamic model of consumer search or chain pricing choices; I leave that to future work and proceed with the single $(\hat{\alpha}_0, \hat{\alpha}_1, \hat{\sigma})$ for the welfare calculation in §5.4, which uses tier-level prices where shares are stable.

A.1 The product-firm demand the structural model inherits

The dynamic game inherits demand at the chain-by-drug level. Each chain i's quality intercept $\hat{\phi}_{ij}$ is recovered from the within-nest shares $\hat{s}^{w}_{i|j}$ observed in the coordinated window via (1) at the common coordinated price; the instrumented coefficient $\hat{\alpha}_j$ governs the price response and the market size N_j is anchored on a normal-period median quantity. Two out-of-sample checks confirm the mapping reproduces both the *market shares* and the *quantities* the game needs.

Market share. Intercepts recovered at the coordinated price predict the within-nest shares in two regimes they were not fit to—the price war and the supra-competitive rent—to a median absolute error of 1.6 percentage points (mean 2.5; 90th percentile 5.6) across the 216 drugs. The coordinated window is in-sample and reproduces exactly.

Quantity. With N_j anchored on the normal-period quantity, the implied three-chain quantity $N_j \sum_i \hat{s}_{ij}$ matches the observed quantity to a median ratio of 0.97, 0.99, and 0.98 at the war, coordinated, and rent tiers (interquartile range $\approx [0.85, 1.13]$; log-correlation 0.75–0.85). Anchoring instead on the absolute-peak quantity—the price-war stocking-up spike—over-states quantity by a third and is rejected.

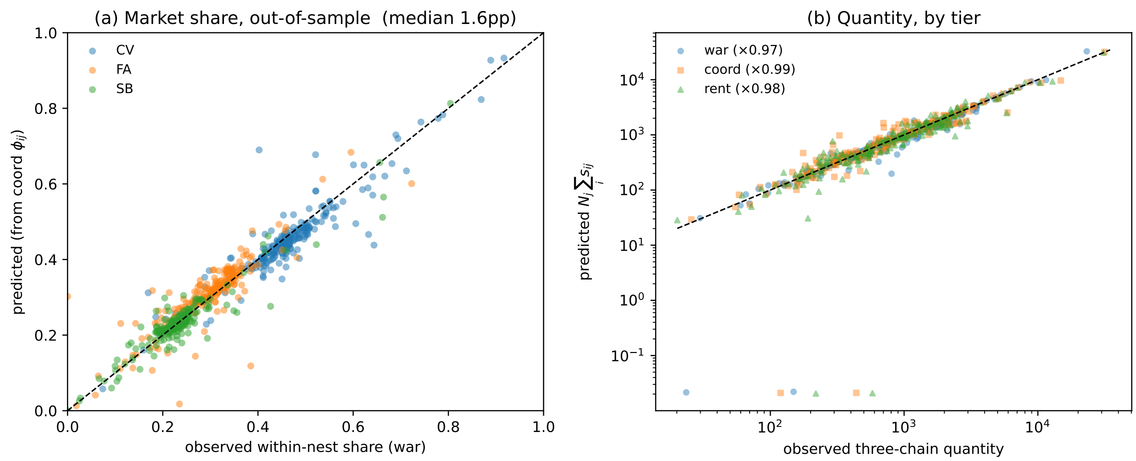
The chain-specific undercut gains reduce to the symmetric gain when the three chains' within-shares are equal and correlate 0.94 with it across drugs, so the product-firm layer nests the symmetric mapping as a special case.

B Hand-coded coordination events vs. the binary cartel indicator

B.1 Event coding versus the binary cartel indicator

The case data include a coordination indicator, $\text{col}_{fjt} \in \{0, 1\}$, equal to one when chain f's price for drug j on day t is in the coordinated (post-increase) regime. This indicator records whether a firm has joined the coordinated price but not how far prices have risen: a $0 \to 1$ flip marks the first coordinated increase, yet the variable cannot distinguish a first increase from a second or third. My hand-coding instead records the full ladder of

Figure 1: Out-of-sample market-share and quantity fit



Note: Panel (a): predicted versus observed within-nest shares in the price war, using intercepts $\hat{\phi}_{ij}$ recovered from the coordinated window (out-of-sample); the 45° line is exact fit. Panel (b): predicted three-chain quantity $N_j \sum_i \hat{s}_{ij}$ versus observed, by tier (log scale). 216 drugs with a complete war / coordinated / rent price triple.

coordinated increases for each drug: every transition from price tier ℓ to $\ell+1$, its date, and the firm that moved first.

Table 3 compares the two. The binary indicator coincides with most first-round onsets but with only 22% of the second- and third-round increases, and in 127 of the 137 multi-round drugs it records fewer steps than my coding. Because the mechanism I study is the sequential, multi-round escalation itself, the tier ladder, not a single on/off flag, is the economically relevant unit of observation. Figure 2 illustrates with four representative drugs: prices climb in discrete plateaus that my coding tracks step by step, while col collapses the entire episode into one block.

Event-time comparison. I further compare the hand-coded event ladder with the FNE official cartel indicator $\text{col}_{ijt} \in \{0, 1\}$ at the drug level. For each drug j I define the FNE official event time as the first day on which $\text{col}_{\text{CV},jt} = \text{col}_{\text{FA},jt} = \text{col}_{\text{SB},jt} = 1$. Of the 205 drugs that have an event in both panels, the median difference (FNE day minus hand-coded day) is -2 days, with inter-quartile range $[-3, +5]$ days and Pearson correlation $r = 0.768$. Seventy-nine percent of drugs are within seven days of each other across the two panels, and 95% are within sixty days.

Table 3: Hand-coded ladder versus binary indicator

	Value
Drugs in the coordination sample	222
Hand-coded coordinated price increases	442
first-round onsets (tier $0 \to 1$)	293
later-round steps (tier $1 \to 2$, $2 \to 3$)	149
Drugs with $\geq$ 2 coordinated increases	137
col $0 \to 1$ flips (drug level)	249
First-round onsets matched by a col flip	188 (64%)
Later-round steps matched by a col flip	33 (22%)
Multi-round drugs where col records fewer steps	127 of 137
Leader agreement (col first-mover vs. hand-coded leader)	57%

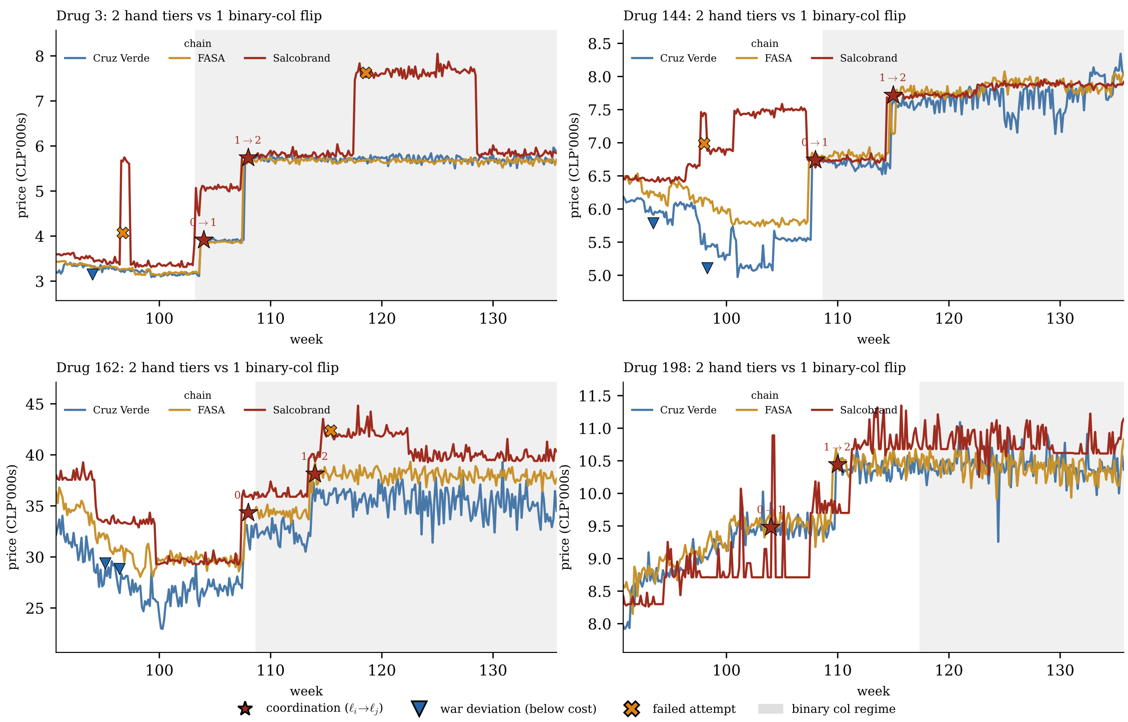
Notes: a col flip is matched to a hand-coded event if the two occur within 15 days. Later-round steps have no counterpart in a binary indicator by construction.

Note: Hand-coded event ladder vs. the binary col indicator

C Additional descriptive tables

This appendix collects the descriptive tables underlying §3: the markup ladder by tier and chain (Table 4), the below-cost exposure and coordination-leadership split (Table 5), the predictors of the coordination sequence (Table 6), and the full coordination ledger by regime window and magnitude (Table 7).

Figure 2: Event taxonomy versus the binary indicator



Note: Each event is marked at its own date and price by nature: coordination tier transitions (stars, labelled $\ell_i \to \ell_j$), below-cost war deviations (triangles), and failed attempts (crosses); post-coordination defections use diamonds where present. Prices move in discrete tiers that the hand-coding records individually, with their nature distinguished, whereas the binary indicator marks only the onset of the coordinated regime.

Table 4: Markups by tier and chain

	Tier 0 (pre-coord.)	Tier 1 (1st coord.)	Tier 2 (2nd coord.)
Panel A. Unweighted mean markup			
Cruz Verde	0.025	0.302	0.596
FASA	0.041	0.304	0.592
Salcobrand	0.047	0.310	0.610
Panel B. Quantity-weighted mean markup			
Cruz Verde	−0.158	0.105	0.371
FASA	−0.154	0.089	0.342
Salcobrand	−0.122	0.127	0.405
Panel C. Count of products with negative markup			
Cruz Verde	133	60	12
FASA	131	59	14
Salcobrand	127	53	11

Note: Panel B: quantity-weighted mean markup. Panel C: count of chain-drug-weeks with negative markup. Markup $m_{ijt} = (p_{ijt} - c_{jt})/p_{ijt}$. Tiers correspond to hand-coded coordination events (§2). N = 222 drugs.

Table 5: Below-cost exposure and coordination leadership

	Cruz Verde	FASA	Salcobrand
Panel A. Pre-ban below-cost exposure			
Fraction below cost	31.3%	26.9%	25.6%
Median $(p - c)$ (10^3 CLP/unit)	0.70	0.87	0.98
Panel B. Coordination leadership, by period (initiating chain)			
War (pre-cartel)	0	0	2
Cartel wave (Dec'07–Mar'08)	14	59	211
Post-investigation rent	36	7	34
All coordinations	**50**	**66**	**247**
of which first-tier ($0 \to 1$)	13	42	165
of which rent ($1 \to 2$)	37	24	82
Share of all (n = 363)	13.8%	18.2%	68.0%

Note: *Panel A* (pre-ban below-cost exposure): drug-weeks with $p_{ijt} < c_{jt}$ in the pre-ban window (weeks 1–94; 9,945 drug-weeks per chain). *Panel B* (coordination leadership): the initiating chain of each of the 363 coordination events—the 220 first coordinations ($0 \to 1$) and the 143 rent steps ($1 \to 2$)—split by the war (pre-cartel), cartel (3 Dec 2007–31 Mar 2008, to the FNE *Oficio*) and post-investigation periods, so the rows reconcile to the full ledger (2 + 284 + 77 = 363). Salcobrand leads the war and the cartel wave; Cruz Verde overtakes it in the post-investigation rent (the SB$\to$CV handoff).

Table 6: Predictors of the coordination sequence

	OLS: first coordination day				Cox PH: HR		
Covariate	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\log r_j$ (market size)	−4.08	−4.83	−4.96*	−4.95*	1.240***	1.236***	1.312***
	(3.31)	(3.12)	(2.24)	(2.36)	(3.06)	(2.94)	(2.86)
$\hat{\alpha}_j$ (elasticity)	−1.46	−3.51	−1.68	−1.94	1.088	1.160†	1.032
	(3.04)	(2.53)	(2.06)	(3.32)	(1.64)	(1.67)	(0.37)
Below-cost gap		1.38	0.09	0.05	1.006	1.020	1.095
		(2.65)	(1.02)	(1.14)	(0.14)	(0.46)	(1.08)
$\hat{\alpha}_j^2$				0.18		0.968	
				(1.60)		(1.14)	
$\hat{\alpha}_j \times \log r_j$				−0.04		1.042	
				(2.42)		(0.52)	
Drug-type FE	No	Yes	Yes	Yes	Yes	Yes	No
Manufacturer FE	No	No	**Yes**	**Yes**	No	No	**Yes**
R^2	0.018	0.087	**0.611**	0.611	—	—	—
Concordance (C)	—	—	—	—	0.594	0.601	0.556
Pseudo-R^2	—	—	—	—	0.017	0.018	0.014

Note: Outcome: the calendar day of the drug's first coordinated increase (OLS, larger = later) or its coordination hazard (Cox, HR > 1 = earlier); continuous regressors standardised. Sample: the coordinated $J = 220$, $N = 219$ after dropping one drug with incomplete pre-ban characteristics. OLS t-statistics (columns 1–4) and Cox z-statistics (columns 5–7) in parentheses. Column (7) is the Cox stratified by manufacturer—a within-laboratory baseline hazard, the counterpart of column 3—for which I report Harrell's concordance (C; 0.5 = chance) and McFadden's pseudo-R^2. OLS standard errors clustered by manufacturer ($G = 37$); Cox standard errors robust. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$, $^{\dagger}p < 0.10$ (one-sided).

Table 7: The coordination ledger by regime window and by magnitude

Panel A. Events by kind and regime window						
Event kind	Pre-decline (2006)	War/dec. (Jan–5Nov'07)	Pre-cartel (6Nov–2Dec'07)	Cartel (3Dec–31Mar)	Post-invest. (1Apr–Dec'08)	Total
First coordination (Tier $0 \to 1$)	0	1	1	216	2	**220**
Rent step (Tier $1 \to 2$)	0	0	0	77	66	**143**
Failed attempt (reverted)	86	99	23	4	67	279
Sustained-unilateral (held)	62	50	12	20	42	186
Total events	**148**	**150**	**36**	**317**	**177**	**828**
Panel B. Coordination steps by tier transition and magnitude						
	Large (≥ 25%)	Medium (15–25%)	Small (12–15%)	< 12%		Total
$0 \to 1$ (first coord.)	177	31	3	9		**220**
$1 \to 2$ (rent step)	51	47	21	24		**143**
All coord. events	**228**	**78**	**24**	**33**		**363**

Note: *Panel A* sub-windows: "Pre-decline" = 2006 (prices flat at ≈ 7.7); "War/decline" = Jan 2007–5 Nov 2007 (the decline begins January 2007 and Cruz Verde's August-2007 comparative-price campaign intensified it); "Pre-cartel" = 6 Nov–2 Dec 2007 (civil-court injunction to the day before the first three-chain success); "Cartel" = 3 Dec 2007–31 Mar 2008 (to the day the FNE opened its investigation, notice No. 419); "Post-investigation" = 1 Apr–Dec 2008. The **220** first coordinations form the cartel in the Dec–Mar window; the rent then deepens through 2008—77 of the **143** rent steps fall in the cartel window, the other 66 post-investigation, as the regime price climbs toward 15. *Panel B* magnitude is the price increase over the prior tier (the war price for the $0 \to 1$ step, the Tier-1 price for the $1 \to 2$ rent): "large" ≥ 25%, "medium" 15–25%, "small" 12–15%, the rest smaller week-aligned steps. Most are large (228 of 363 steps are ≥ 25%). Coordinations and rents are from the dynamic ledger (which tracks the rent past the hand-coding's April cutoff); failed lifts and sustained-unilateral holds are the hand-coded series.

D Market-size robustness

The demand estimator of Section 4 uses $N_j = \max_t Q^{\text{wk}}_{jt}/\hat{\rho} + 1$ with $\hat{\rho} = 0.92$ (three-chain market share, FNE 2008). I re-estimated the model under two alternative calibrations.

1. **Higher chain-share estimate.** Set $\hat{\rho} = 0.95$ (Díaz and Galetovic, 2017). This lowers N_j by 3.3% and raises the outside share at the peak week from 8% to 5%. Point estimates of $(\hat{\alpha}, \hat{\sigma})$ shift by less than ±0.005.
2. **Constant national population.** Set $N_j = N^*/J$ for all j, with $N^* = 4 \times 10^6$ (Chile's adult prescription-buying population). $(\hat{\alpha}, \hat{\sigma})$ shift by less than ±0.01.

The own-price elasticity dispersion across drugs is essentially unchanged across the three specifications, confirming that the chain-by-drug fixed effect ϕ_{ij} absorbs level mis-scaling of N_j.

E Data construction

The transaction data, obtained from the FNE under Chile's transparency law (Ley No. 20.285), cover all purchases of the 222 case products from the three chains over 2006–2008; each record carries the drug, chain, date and time, list price, transacted price, and units. I aggregate to weekly revenue-weighted averages: daily data are noisy at low volumes, while monthly aggregation would blur the 2–4-day coordination windows. Product attributes (active ingredient and ATC class; package size, pill count, brand-vs-generic status, prescription status) were hand-compiled and cross-checked from three public sources (an internal catalog, DrugBank, and the Chile-specific listing Farmazon) with the small share of conflicts (< 3%) resolved in favour of the Farmazon listing. Wholesale costs are Salcobrand's wholesale prices for November 2007–May 2008, submitted to the Competition Tribunal as expert evidence [Tribunal de Defensa de la Libre Competencia (TDLC), 2012]. For chain-product-periods with no recorded sales I use the chain-product median price; periods with zero quantity are treated as missing, not as a zero price; and within-chain coordination statistics use only chains that stock the product.

Mapping to the 222*-drug TDLC list.* Of the 222 drugs named in the TDLC *Sentencia*, 220 (99.1%) show at least one successful three-chain coordination in the event panel; 211 clear the 15% hand-coding threshold, and 9 are recovered only by the more permissive daily-panel detector or the later-window pass (small tier-0 $\rightarrow$ 1 events of 12–15%). The remaining two, Losopil (Mediapharm) and Progyluton (Bayer), show only single-firm unilateral lifts of 12–14%, with no event in which all three chains follow. Both count as coordinated in the official `col_ijt` indicator—a binary flip covering the full 222-drug list by construction—but my finer panel, which separates single-firm attempts from three-chain agreements, does not classify them as coordinated. The implied three-chain coordination rate on the TDLC list is 220/222 $\approx$ 99.1%.

F The structural estimation sample

The dynamic model of §5 is estimated on the $J = 220$ drugs that complete the war-to-coordination price transition the model is built around. A drug enters the structural panel only if the event ledger records all three of (i) a war/deviation price (a pronounced dip during the price-war window), (ii) a Tier-1 coordinated price, and (iii) an imputed cost; the Tier-2 rent price is imputed when absent, and the per-drug price coefficient α_j falls back to its type median when the demand regression is uninformative.

An initial pass over the hand-coded ledger excluded fifteen drugs. Cross-checking each against its raw price path showed that nine of them do complete the transition, a marked dip during the war window followed by a sustained higher level, that the hand-coding had failed to record; I reclassified these (restoring the missing war and/or Tier-1 events), raising the sample from 207 to 216. A second pass then recovers four more (Table 9): Carvedilol, Chlorpheniramine Maleate, Ethinyl Estradiol, and Oxolamine Citrate are all hand-coded three-chain coordinations—Figure 3 shows each stepping up to a sustained higher level in 2008—whose only missing ledger leg is a flagged war-deviation price. I impute that leg from the drug's demand-panel war-window median (at the 0.82 ratio that aligns the demand-window median with the recorded war-deviation price across the in-sample drugs; `full_structural.build_panel`), which brings the structural sample to $J = 220$—exactly the descriptive count of coordinated drugs in Table 7. This price-path audit doubles as a completeness check on the hand-coding. Two case drugs remain excluded: the Neuractin presentation of Valproic Acid (#202) coordinates in early 2008 but then defects after a monotonic price rise, with no preceding loss-leader war to identify the transition; and Estradiol Valerate/Norgestrel (#215) never reaches a sustained Tier-1 level. Excluding these two is conservative for the coordination-timing moments.

Table 8 lists the moments the SMM criterion targets and the parameter each primarily identifies, the mapping summarised in §5.1.

Table 8: Targeted moments and what they identify

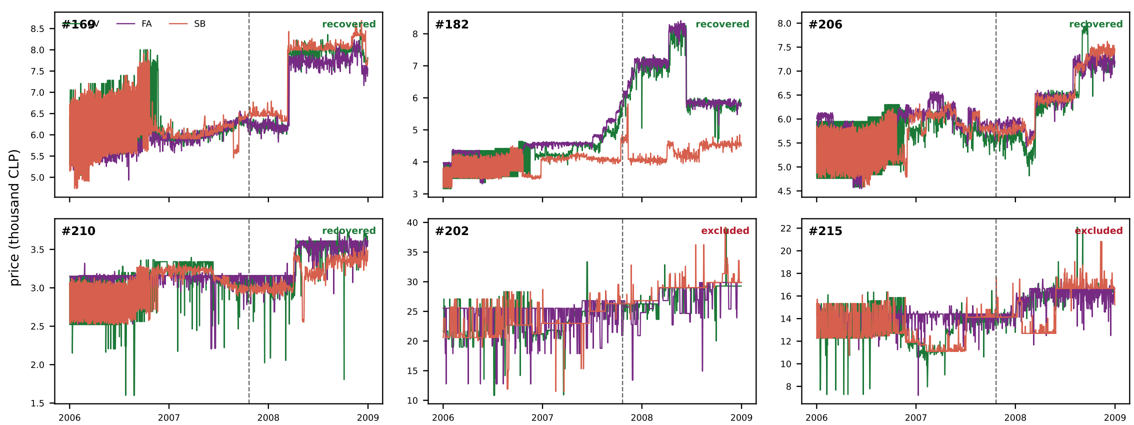
Group	Moment	Primarily identifies
Counts	first ($0 \to 1$) and second ($1 \to 2$) coordination totals	λ; rent penalty χ
Timing	onset week, peak week, wave SD	t_0, λ
	$1 \to 2$ lag behind $0 \to 1$	rent penalty χ
Laboratory	lab-wave SD; labs coordinated by wk. 104; within-lab SD	reach-out λ, τ_e
	corr(log lab size, coordination week)	reach-out λ, τ_e
War floor	below-cost war markup	μ (joint)
War series	weekly below-cost deviations: pre / mid / late / post	g
Micro	corr(pre-ban cheat freq., G_j^{01})	benefit-ordered war
	α_j/margin gap: rent-takers vs. not	rent selectivity
Belief	failed attempts by period (war high, cartel ≈ 0, post rebound)	war prior (a_0, b_0), signal ζ, scrutiny κ
Rent	rent steps by period (in-cartel + post-cartel deepening)	rent penalty χ, scrutiny κ
Leadership	leader share by period (SB wave, CV late rent)	μ_i spread (exposure-corroborated)

Table 9: Borderline case drugs: recovered or excluded

Drug ID	Active ingredient	Type	Status	Ledger note
169	Carvedilol	Chronic OTC	Recovered	war price imputed
182	Chlorpheniramine Maleate	Acute OTC	Recovered	all legs imputed
206	Ethinyl Estradiol	Acute Rx	Recovered	war price imputed
210	Oxolamine Citrate	Acute OTC	Recovered	war price imputed
202	Valproic Acid (Neuractin)	Chronic Rx	Excluded	coordinates, then defects; no war
215	Estradiol Valerate/Norgestrel	Chronic Rx	Excluded	never reaches Tier-1

Note: Four of the six are recovered into the structural sample—each is a hand-coded three-chain coordination whose only missing ledger leg is a war-deviation price, imputed from the demand panel—raising J from 216 to 220, the full count of coordinated drugs; two remain excluded. "Type" is the chronic/acute × prescription/OTC classification.

Figure 3: Six borderline drugs: four recovered, two excluded



Note: All six borderline drugs I adjudicated are plotted—the four recovered (green) I admit to the sample and the two excluded (red) I drop, so only the two red drugs leave the sample (the figure shows both decisions, not only the drops). Each panel plots the three chains' daily price (thousand CLP, cleaned of abnormal jumps as in the demand panel) for one drug; the dashed line marks the advertising ban. The four recovered (green) each step up to a sustained higher level in 2008—a hand-coded three-chain coordination—but their war-period dip is shallow or unrecorded, so I impute the war-deviation price from the demand panel and admit them, taking the structural sample to $J = 220$. The two labelled excluded (red) stay out: #202 rises monotonically and then defects without a preceding loss-leader war, and #215 never reaches a sustained Tier-1 level.

G Robustness to the demand specification

The structural estimate is robust to how the demand side is specified. Table 10 re-estimates the model under three demand specifications on the identical headline criterion: the heterogeneous per-drug $\hat{\alpha}_j$ from the OLS Panel B (*Het. OLS*); the same heterogeneous $\hat{\alpha}_j$ from the rival-price IV Panel B (*Het. IV*, Table 2); and a homogeneous demand that gives every drug the median $\hat{\alpha}$. Both the store-traffic level ($\bar{\mu} \approx 9.3$) and the leadership ordering—Salcobrand leading the cartel wave (SB-share 0.72–0.77) and Cruz Verde taking over the post-investigation rent—are essentially unchanged across all three. The two heterogeneous specifications fit comparably (loss 16.9 and 15.9); homogeneous demand fits worse (loss 30.9) because, without the per-drug heterogeneity, the wave is too concentrated (SD 3.5 vs. the data's 4.6 weeks) and the rent too slow ($1 \to 2$ lag 10 vs. 8)—so the per-drug heterogeneity is what spreads the wave and times the rent.

Table 10: Structural estimates across three demand specifications

	Data	Het. OLS	Het. IV	Homog.
Panel A. Estimated parameters (SMM, het-μ)				
$\bar{\mu}$ mean store-traffic value		9.33	9.48	9.11
λ reach-out scale		0.61	0.64	0.74
τ_e reach-out temperature		168	171	172
t_0 organization onset (wk)		103	103	103
a_0, b_0 belief prior		$1.61, 96$	$2.12, 98$	$2.05, 94$
ζ ad-ban pseudo-successes		229	198	289
χ rent-tier discount		1.96	1.89	2.14
κ FNE-scrutiny on the rent		1.97	1.86	2.56
g Aug. campaign strength		0.72	0.65	0.63
SMM loss		**15.9**	**16.9**	**30.9**
Panel B. Fit on selected moments				
First coordinations	220	211	210	187
Rent steps	143	134	137	**156**
Failed war attempts	123	119	143	141
Cartel successes	284	285	278	270
Post-late successes	49	31	34	44
Peak week	107	106	107	105
Wave SD (weeks)	4.6	4.0	4.4	**3.5**
$1 \to 2$ lag (weeks)	8.0	8.6	8.6	**10.4**
Between-lab var. share	0.59	0.55	0.55	0.52
Spearman(size, week)	−0.30	−0.24	−0.23	−0.23
SB leads cartel wave	0.74	0.72	0.72	0.77
CV leads post-invest. rent	0.65	0.64	0.65	0.94

Note: each column re-estimates the twelve endogenous-lab parameters (the three chain-specific μ_i and the nine baseline) by SMM on the headline criterion; the engagement window $w = 26$, δ, τ and the holiday factor are fixed. *Het. OLS* and *Het. IV* differ only in the per-drug $\hat{\alpha}_j$ (Table 2, Panel B: OLS vs. rival-price 2SLS); *Homog.* sets every $\hat{\alpha}_j$ to the median and selects rent-eligible drugs by the flow condition alone. Panel A reports the mean store traffic $\bar{\mu}$; the re-estimated μ_i spread reproduces the leadership in every column (the SB-leads row). The main text reports the rival-IV estimate; the mechanism—belief transition, lab-organized wave, and the Salcobrand→Cruz Verde leadership handoff—and the store-traffic spread that carries it survive all three demands.

H Structural robustness checks

Each check switches off one ingredient of the model of §5 and re-simulates, reporting the same fit moments as Table 4 (Panel B). Table 11 collects them and Figures 4–5 show the paths.

RC1: the ban shifts beliefs but does not collapse price sensitivity. Holding post-ban demand equal to pre-ban demand, the advertising restriction still delivers its public belief jump but the static undercut gain G^{01} does *not* flip. The first wave still forms—the belief carries it—but the coordination is no longer self-enforcing: the rent step collapses (56 against the data's 143) and post-cartel failed lifts flood (152 against 46; Figure 4B), because a high belief alone cannot hold a price that is not a static best response. The elasticity collapse is what makes the coordinated price stick.

RC2: no store traffic ($\mu = 0$). With $\mu = 0$ there is no cross-category profit from being cheapest, so the symmetric Bertrand–Nash war floor is a +56% markup—*above* cost—against the observed below-cost markup of −10%. The defining feature of the pre-ban data, sustained below-cost pricing, is then not an equilibrium at all: the loss-leader war cannot be rationalised without a positive store-traffic value.

RC3: leadership requires the store-traffic spread. Leadership is driven by the spread in the chain-specific store-traffic values μ_i (§5.1). With the estimated spread Salcobrand leads 72% of the cartel wave (record 74%); flattening it to a common μ—the same war floor, no chain difference—collapses leadership to chance ($\sim 1/3$ each, Figure 5). The μ_i spread alone selects the Salcobrand-first order.

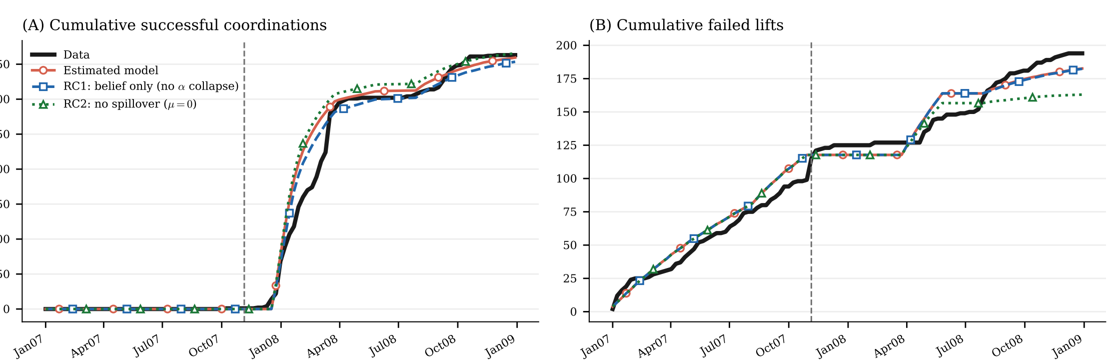
RC4: an outside option in place of chain-specific store traffic. The headline carries leadership with a chain-specific store-traffic value μ_i (§5.1). The alternative the Editor first proposed places the asymmetry on a firm-side outside option instead: a common μ for all chains plus three insulation parameters ψ_i that cushion a fraction of each chain's below-cost loss, ψ read from the chains' reliance on Chilean retail ($\psi_{\text{SB}} = 0$ for the purely-domestic Salcobrand, $\psi_{\text{FA}}, \psi_{\text{CV}} > 0$ from the rivals' diversification). With the reliance-based $\psi = (0.90, 0.76, 0)$ the model still produces the Salcobrand-first ordering—Salcobrand leads 56% of the cartel wave (data 74%)—but fits less well than the store-traffic spread (loss 20.5 against the headline's 16.9, where Salcobrand leads 72%). The two channels are alike in kind—both make the below-cost war differentially painful, a lower store-traffic value and a thinner outside option each raising a chain's escape value—but here the store-traffic spread both fits better and reproduces the leadership more sharply. I take the store-traffic spread as the headline for three reasons. It is the *same* primitive that already rationalises the war, so the war behaviour and the leadership follow from one object. Its ordering is corroborated out of sample by the below-cost exposure (Table 5: $\hat{\mu}_{\text{CV}} > \hat{\mu}_{\text{FA}} > \hat{\mu}_{\text{SB}}$ matches the exposure exactly), whereas the ψ_i magnitudes—in particular $\psi_{\text{CV}} = 0.90$, asserting Cruz Verde is 90% insulated—are pinned by no chain financial. And the heterogeneity it asks for is in the store-wide basket—a distinct object from the chain-by-drug demand quality $\hat{\phi}_{ij}$, which the demand data pin down out of sample (§4). The ψ column of Table 11 reports the fit.

Table 11: Structural robustness checks: fit moments

Moment	Data	Estimated	RC1 belief	RC2 $\mu = 0$	RC3 uniform-μ	RC4 ψ
First coordinations ($0 \to 1$)	220	206	200	220	220	220
Rent steps ($1 \to 2$)	143	137	137	145	140	134
Below-cost war markup	−0.10	−0.10	−0.10	**+0.56**	−0.10	−0.10
Successes: cartel	284	283	270	309	299	283
Failed lifts: war	123	120	120	120	119	110
Failed lifts: cartel	4	0	0	0	0	0
Failed lifts: post-early	46	50	53	43	52	51
corr(log lab size, week)	−0.29	−0.44	−0.38	−0.43	−0.45	−0.39
SB share of cartel-wave leads	0.74	0.72	0.73	**0.33**	**0.33**	0.56

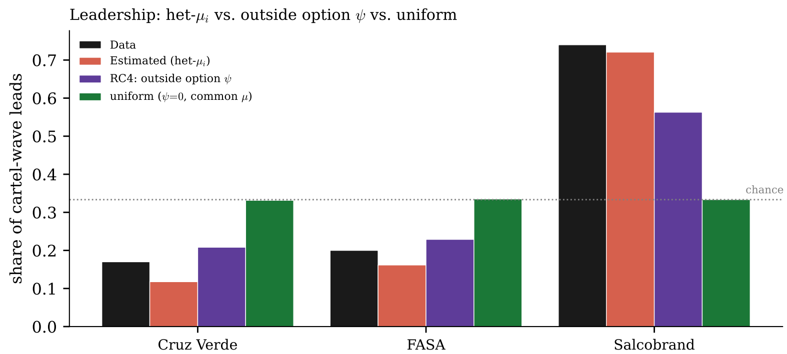
Note: each column re-simulates the headline model ($B = 200$) with one ingredient switched off—RC1: post-ban demand set equal to pre-ban (no elasticity collapse), which collapses the rent and floods post-cartel failures; RC2: $\mu = 0$ (no store traffic), which turns the war floor positive and, with no μ_i spread, drops leadership to chance; RC3: a uniform store-traffic value (same war floor, no spread), which keeps the floor but still drops leadership to chance, so the *spread* is what carries it. RC4 instead re-estimates the leadership with a common μ and the reliance-based outside option $\psi = (0.90, 0.76, 0)$: it produces Salcobrand's lead (0.56, data 0.74) at loss 20.5 against the headline's 16.9, fitting less well than the store-traffic spread. Compare the "Estimated" column to Table 4, Panel B.

Figure 4: Robustness: the wave and the failed lifts



Note: cumulative successful coordinations (A) and failed lifts (B), model specifications versus data. RC1 (belief only, no α collapse) reproduces the wave but floods with post-ban failed lifts; the estimated model holds clean. Vertical line: the November ban.

Figure 5: Leadership: store-traffic spread vs. the ψ alternative



Note: share of cartel-wave coordinations each chain leads. The estimated store-traffic spread (het-μ) puts Salcobrand first, matching the record; the outside-option ψ alternative (RC4) gives the near-identical split; flattening the spread to a uniform μ drops leadership to chance (1/3).

Robustness of the coordination flip. Table 12 reports the threshold cushion behind the parameter-free flip discussed in §5.3. Panel A varies the size of the post-ban drop in α; Panel B perturbs the wholesale-cost and nesting calibration. The band $[\mu^*_{\text{pre}}, \mu^*_{\text{post}}]$ never closes.

Table 12: Robustness of the coordination flip

Panel A. Cushion vs. the size of the α drop			
post-ban α	μ^*_{post}	cushion $\mu^*_{\text{post}}/\mu^*_{\text{pre}}$	band
0.029 (cartel-excl. est.)	30.3	3.5×	open
0.060	14.6	1.7×	open
0.100 (rival-IV upper bound)	8.8	1.0×	open (sliver)
Panel B. Sensitivity of $[\mu^*_{\text{pre}}, \mu^*_{\text{post}}]$ to the calibration			
perturbation	μ^*_{pre}	μ^*_{post}	ratio
baseline	8.6	30.3	3.5×
wholesale cost −10%	7.9	29.6	3.8×
wholesale cost +10%	9.3	31.0	3.3×
nesting $\sigma = 0.30$	9.8	34.8	3.6×
nesting $\sigma = 0.50$	7.2	25.0	3.5×

Note: $\mu^*(\alpha)$ is the store-traffic value at which undercutting the cartel ties coordinating; the flip requires $\mu^*_{\text{pre}} < \mu < \mu^*_{\text{post}}$. Panel A varies the size of the post-ban drop in α ($\mu^*_{\text{pre}} = 8.6$ fixed at $\alpha = 0.103$); Panel B perturbs the calibration.

Belief and payoff specifications. Table 13 re-estimates the model with each belief channel and each payoff ingredient switched off in turn, the evidence behind §5.3 and the per-moment companion to Figure 7. The focal jump dominates: dropping the Bayesian learning—jump-only—still reproduces the cartel wave and fits essentially as well as the headline (loss 19.3 against 16.9), so the event-tally learning is a minor refinement. The focal jump itself is what ignites the wave: with it switched off ($\zeta = 0$, learning-only) the coordination never fires in-sample—a degenerate corner (0 coordinations, loss $\sim$2300)—so the commonly observed ban, not the slow tally, is what selects the coordinated equilibrium. The outside option's separate role—selecting who leads—is the re-simulated check in Table 11 (RC3: without it leadership collapses to uniform).

Table 13: Structural specifications versus the data: per-moment fit

Moment	Data	Estimated	No belief	Jump only	Learn only	Belief only	$\mu = 0$
First coordinations ($0 \to 1$)	220	206	220	212	0	192	220
Rent steps ($1 \to 2$)	143	137	144	139	0	53	138
Peak week	107	106	52	106	—	104	107
Wave spread (SD, weeks)	4.6	4.1	7.7	4.7	—	4.6	4.0
Coordination successes							
war	2	0	**217**	0	0	0	0
cartel	284	283	112	281	0	207	286
Failed lifts, war	123	120	177	106	127	111	120
corr(log lab size, week)	−0.29	−0.44	−0.48	−0.42	—	−0.32	−0.42
SB leads the cartel wave	0.74	0.72	0.94	0.72	—	0.67	**0.34**
SMM criterion (loss)	—	**16.9**	**3433**	**19.3**	**∼2300**	**284**	**548**

Note: per-moment fit behind Figure 7; this is the single re-estimated-specification table, each column re-estimated under the identical SMM criterion with the endogenous-lab engagement (the *re-simulated* ceteris-paribus checks are Table 11). "Belief only" shifts beliefs at the ban but holds price sensitivity at its pre-ban level (so G^{01} never flips); $\mu = 0$ removes the store-traffic floor. No belief floods the war (217 pre-ban coordinations vs the data's 2); learn-only never ignites the wave; both payoff knockouts mis-shape the wave. The focal ad-ban jump dominates: with it switched off (learning-only, $\zeta = 0$) the wave never fires in-sample (a degenerate corner, 0 coordinations, loss ∼2300), while the Bayesian event-learning is a minor refinement (jump-only, which drops it, still reproduces the wave and fits essentially as well, loss 19.3 against the headline's 16.9).

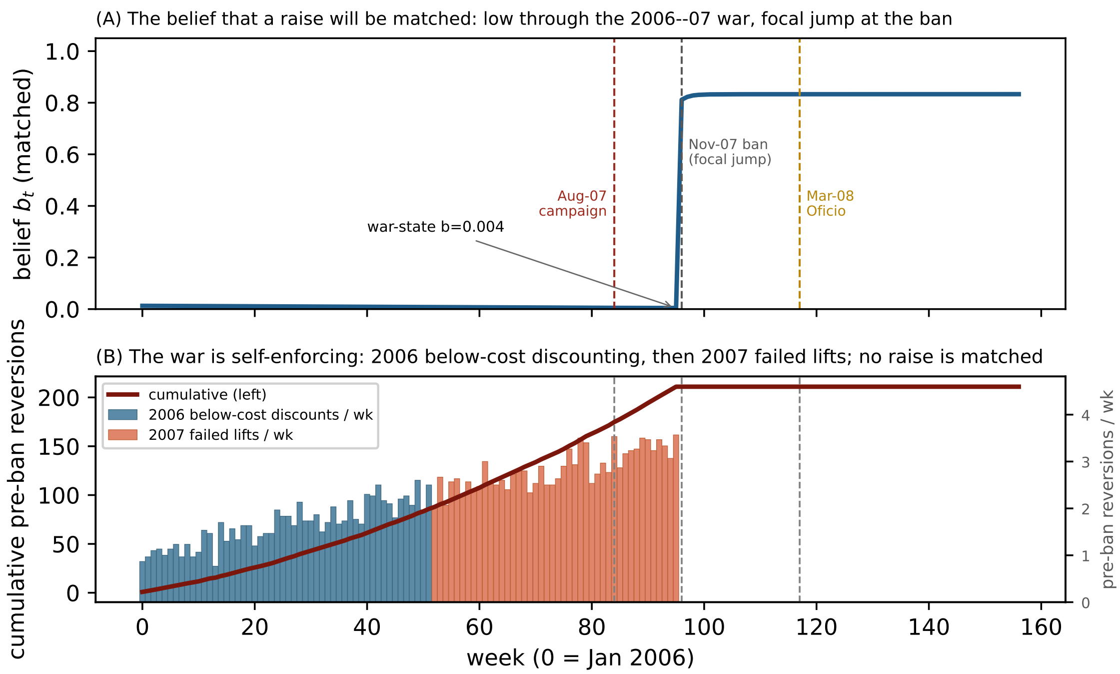
H.1 The war forms endogenously, and experimentation cannot escape it

Table 13 shows the focal jump is essential and the Bayesian learning a contributing refinement. To see why, I roll the belief and price dynamics from January 2006—a year before the structural window—through the war and into the cartel (Figure 6). The war is the low-belief one of two coexisting equilibria—high-belief coordination is dynamically sustainable throughout (§5.3), so the question is never whether the chains *can* coordinate but which equilibrium is selected—and it forms on its own. Through 2006 the war is fought by below-cost discounting: a chain cuts to steal store traffic, the rivals match within days, and the cut springs back (Fact 1). This *is* the war, and no chain yet holds a coordinated increase. As the below-cost pain deepens into 2007, clean attempts to lift out of it appear; but pre-ban every lift is undercut ($G_j^{01} > 0$) and reverts, so none sticks. The simulation reproduces the record on both counts—about 86 below-cost reversions in 2006 and 125 failed lifts in 2007, against 87 and 123, with zero successes. Discount or failed lift, every pre-ban reversion teaches the same lesson, that a raise will not be matched, so the belief never climbs above its low prior: it drifts from 0.013 to 0.004 by the ban, and the pessimism is self-fulfilling. The 6 November ban then delivers the focal jump and flips $G_j^{01} < 0$; the belief leaps to 0.81 and the cartel forms.

What the focal jump adds: ignition on the observed timescale. Suppose the ban's payoff effect occurs but there is no focal jump ($\zeta = 0$), so the chains raise the belief through the tally alone. Post-ban the static flip ($G_j^{01} < 0$) makes a coordinated raise a best response for the static-Nash majority, and a co-led raise faces only the b_t^{3-k} gate for k simultaneous leaders, far weaker than the single-leader b_t^2—so coordination is not *logically* impossible. But it is far too slow to explain the record: re-estimated without the jump, the wave *never ignites within the sample*—no coordinated wave forms by the window's end, fitting an order of magnitude worse ($\sim$2300 against the headline 16.9). By pure experimentation the static-Nash majority would coordinate only over a horizon far longer than the observed window; its exact length turns on the learning rate, which the data identify only loosely, but on any reading it is far beyond the four-month wave, and the most-elastic drugs, whose G_j^{01} stays positive even post-ban, never coordinate without the belief at all. The focal jump's role is therefore to make the observed wave *possible on the observed timescale*: the commonly observed event resolves the higher-order belief in one step, compressing what bare experimentation would reach only much later into the sharp December onset the record shows.

The low belief is disciplined out of sample. The pessimism is not fit to the estimation window. Through 2006 the chains fought a deepening below-cost war and *never* sustained a coordinated increase—an observed coordination frequency of 0.00, against the estimated prior $a_0/(a_0 + b_0) = 0.013$; folding the 2006 record into the tally only lowers it. By the time the structural window opens, no raise had ever been matched.

Figure 6: The war forms endogenously from the belief dynamics

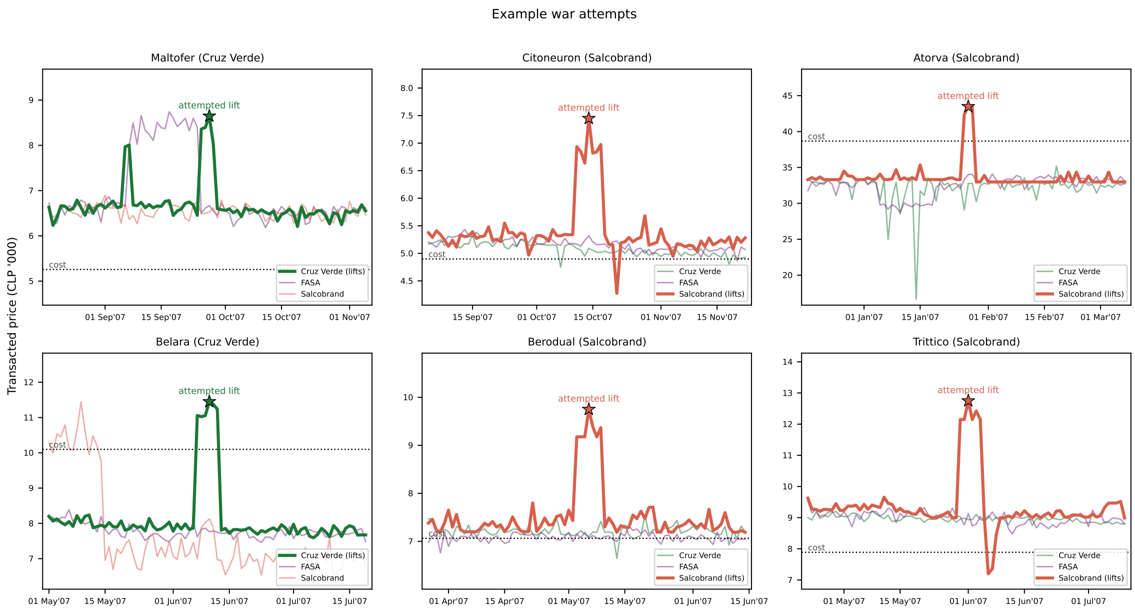


Note: belief and price dynamics rolled from January 2006 (week 0). **(A)** the belief that a raise will be matched: low and self-fulfilling through the 2006–07 war, then the focal jump at the 6 November ban. **(B)** the pre-ban price reversions that keep it low: 2006 is below-cost discounting (the war itself, Fact 1), 2007 the clean failed coordination lifts; the simulation matches the record (86/125 versus 87/123) with zero pre-ban successes. The reversion rate is calibrated to the two annual totals; the belief, hold and jump are the estimated model.

I Example war attempts

Figure 7 shows six representative war attempts from the pre-ban period: episodes in which one chain raised a drug's price toward a coordinated level, the two rivals did not follow but undercut it, and the lift reverted to the below-cost war floor within weeks. These are the ledger's clean `failed_attempt` events (101 lift-then-revert episodes in the January–November 2007 war window, after setting aside 34 lower-amplitude price wobbles that do not clear a rise-and-revert threshold relative to the drug's own volatility), which the structural model reproduces through its belief-driven, sporadic war lifts (§5; 100 simulated). Salcobrand is the lifter in four of the six (Cruz Verde in the other two), consistent with its role as the coordination leader.

Figure 7: Six example war attempts



Note: In each panel the heavy line is the chain that raised price (marked ★ at the lift); the two light lines are the rivals, which stay near or below cost (dotted line) and undercut the lift, so it reverts within weeks. Transacted prices, three chains, ±6-week window around the event.

J Regulatory timeline and case outcome

The episode's documented chronology: a comparative-price advertising campaign by Cruz Verde (August 2007); the CONAR self-regulatory ruling (CONAR 704/07, 7 September 2007) and the binding Civil Court *medida precautoria* prohibiting comparative price advertising (6 November 2007); the first three-chain coordinated increase (3 December 2007) and the wave running through April 2008; the FNE's complaint (Requerimiento, Rol C No. 184-08, 2008; Fiscalía Nacional Económica (FNE), 2008); and the Competition Tribunal's conviction on 206 drugs with fines of about US\$38M (Sentencia No. 119/2012; Tribunal de Defensa de la Libre Competencia (TDLC), 2012), upheld by the Supreme Court (Rol No. 2578-2012). Civil compensation followed: Cruz Verde and Salcobrand settled in November 2020 (CLP 1.1bn to ~53,000 consumers) and FASA in August 2025 (CLP 980M to ~34,000), for total consumer compensation exceeding CLP 2.08bn (~US\$3.9M) across more than 87,000 consumers (SERNAC settlement records). These realized magnitudes are an external check on the welfare accounting of §5.4.

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